Smart India Hackathon 2026

Idea Screening & Verification Report · prepared 24 August 2026 · Team of 6

Bottom line

All **226 problem statements** were screened. **8** reached final verification. **2 clear every criterion**, and **4 more clear once the novelty requirement is set aside** (novelty is something we can engineer in; missing data is not).

Recommended: SIH26011 (3D ULPIN) — its core calculation has been *proven working on real government data*, not just argued on paper. **Strong second: substandard-drug batch alerts** under an AICTE open slot.

1. How we got from 226 to 8

Stage	Count	What happened
All problem statements	226	172 software · 54 hardware · 18 themes
Ministry / company PS	192	Every one screened and given a written verdict
AICTE open slots	34	No fixed title — we invent the idea. 12 software themes worked
Reached shortlist	8	Survived competitor sweeps and criteria screening
Passed hard verification	**2**	**Data opened and tested by hand, not just searched**

Why the survival rate is low. Nine separate ideas died to a government product that already exists — myScheme, MAITRI 2.0, GeoMGNREGA, GeM Startup Runway, panchayat-level weather forecasts, C-Flood, biomedical-waste barcode tracking, YES-TECH, and the unified unclaimed-assets portal. Ministries write problem statements about programmes they are already funding. That is a structural feature of SIH, not bad luck.

2. How ideas were scored

Seven axes, each out of 10. Overall score is the plain average, so anyone can recheck it. **Any single axis at 4 or below blocks the idea —** a fatal weakness must never hide behind a good average.

Axis	What it asks
Problem	Is the problem real and measured by a credible published source?
User	Is the target user precise, nameable and countable — not "farmers in India"?
Demo	Is there something visible a judge can watch working live?
Judge	Can a non-expert judge grasp problem and cleverness in 60 seconds?
Novelty	Scored *after* a competitor sweep, never from memory or assumption
Feasibility	Can *we* actually obtain the data and build it?
Impact	Scale × severity, tied to the precise user group above

3. Master scorecard — all 8 finalists

PS No.	Idea	Ministry	Prob	User	Demo	Judge	Novel	Feas	Imp	Score	Status
SIH26011	**3D ULPIN — land record for every flat**	Rural Development	8	9	9	9	6	7	8	**8.0**	PASS
SIH26196	**Substandard-drug batch alerts** (AICTE open slot)	AICTE — MedTech	8	8	8	9	6	7	8	**7.7**	PASS
SIH26001	Landslide early warning, North East	MDoNER	9	7	8	8	3	6	9	**7.1**	NOVELTY
SIH26100	GeM tender compliance checking	Petroleum & Natural Gas	7	8	6	6	6	8	7	**6.9**	BELOW BAR
SIH26066	OceanEmbed — subsurface ocean temperature	Earth Sciences	7	7	8	7	3	9	6	**6.7**	NOVELTY
SIH26162	Industrial fire & thermal source detection	NTRO	6	6	8	7	4	9	6	**6.6**	NOVELTY
SIH26102	MPLAD fund anomaly detection	MoSPI	8	8	5	8	4	3	7	—	DEAD
SIH26034	Packaged-goods label compliance	Consumer Affairs	6	8	7	6	5	4	5	—	DEAD

Red cells mark a blocking score (4 or below). Note: SIH26011 is tagged "Space Technology" on the SIH portal — that is a data-entry error on their side; it is a land-records problem statement.

4. Second pass — with novelty switched off

Three ideas failed *only* on novelty. Since we can design novelty into a solution but cannot conjure data that does not exist, they were re-scored on the remaining six axes. MPLAD and Legal Metrology are excluded — they failed on **data**, which no amount of clever design repairs.

PS No.	Idea	Prob	User	Demo	Judge	Feas	Imp	Score
SIH26011	**3D ULPIN**	8	9	9	9	7	8	**8.3**
SIH26196	**Drug batch alerts**	8	8	8	9	7	8	**8.0**
SIH26001	Landslide warning, NER	9	7	8	8	6	9	**7.8**
SIH26066	OceanEmbed	7	7	8	7	9	6	**7.3**
SIH26100	GeM compliance	7	8	6	6	8	7	**7.0**
SIH26162	Industrial fires	6	6	8	7	9	6	**7.0**

5. First choice — SIH26011, 3D ULPIN

Problem statement	**3D ULPIN Generation and Vertical Property Mapping System**
Ministry	Ministry of Rural Development · Software
Score	**8.0** with novelty · **8.3** without

The problem in one line

Land records in India were designed for fields, not towers. If you own a flat, the land under your building is recorded in the *builder*'s or the *society*'s name — you have no documented share of it. Maharashtra made the **Vertical Property Card mandatory from 1 January 2026**, and existing societies can apply until December 2027. But for older buildings, someone has to actually compute every flat's land share, and nobody has automated it.

Why this is the strongest option: we proved the maths works

We did not merely argue this is feasible. We opened MahaRERA's public portal, pulled a real registered project — **P50500000005, "GREEN CITY 3", Nagpur** — and computed the official land share for every unit in it.

```
Units in project        : 232
Total carpet area       : 10,570.88 sq.m
Total land area         :  8,120.96 sq.m

Example results
  52.49 sq.m flat  →  40.32 sq.m of land
  43.12 sq.m flat  →  33.13 sq.m of land
  12.09 sq.m shop  →   9.29 sq.m of land

Sum of all 232 land shares :  8,120.96 sq.m
Reconciliation error       :  0.000000 sq.m
```

We then independently confirmed that the official Maharashtra method *is* this formula — land share proportional to carpet area over total carpet area, held as an undivided share. **Our calculation matches the government's own method exactly.**

What the public data gives us — verified on screen

- Total land area of the approved layout, and sanctioned built-up area
- Every building/wing with its sanctioned floor count
- Every unit type with carpet area in sq.m and how many of them exist
- GPS coordinates, survey/CTS number, and boundary descriptions
- Document library including Layout Approval and Building Plan Approval

Limits we will state ourselves, before a judge finds them

- **CAPTCHA on each project page.** Fine for a demo and for society-by-society use; it does block mass automation.
- **RERA began in 2017.** Older buildings are not on it — and those are exactly the societies the scheme most needs. Their plans sit with municipal bodies, obtainable one at a time via RTI.
- Document quality varies; some builders upload the same PDF into several slots.

6. Second choice — substandard-drug batch alerts (AICTE slot SIH26196)

Slot	AICTE Student Innovation · MedTech / BioTech / HealthTech · Software
Score	**7.7** with novelty · **8.0** without

The pitch

"Seventy bad medicine batches are found every month. Nobody tells your chemist. We do."

Verified data — we downloaded it

- CDSCO's March-2025 alert contains **70 flagged batches in that month alone**, and parses cleanly to structured text
- Every record carries the **batch number**, product, manufacturer with full address, expiry date and the exact test it failed
- Real examples: Pantoprazole batch `SP240165` (dissolution failure), Nitrazepam batch `AXI23003P` flagged **spurious**
- The live searchable database covers **2019 to 2026**, currently updated to July 2026, with a **separate tab for fake drugs**

Batch numbers being present was the make-or-break question for this idea. They are present.

What we would build

Scan a medicine strip → read the batch number → check it against eight years of government records → tell the user instantly if that exact batch was flagged. Plus a dashboard of repeat offenders across those eight years.

Honest limitation

The strongest feature — "*your shop is currently holding 40 strips of a recalled batch*" — needs pharmacies to share their stock list. The consumer scan-a-strip version needs no cooperation from anyone and still works.

7. The two that are permanently dead — and why it is not about novelty

SIH26102 — MPLAD fund monitoring

The idea was to check MP-funded works against satellite imagery, since the CAG found works that do not physically exist. We opened the eSAKSHI portal. It publishes **only MP-level allocation** — state, MP name, constituency, amount. **There are no individual work records and no locations.** Without work locations there is nothing to verify from space. Data that does not exist cannot be designed around.

SIH26034 — Packaged-goods label compliance

There is **no public dataset of product labels**. We would have to photograph packets ourselves or scrape retail sites against their terms. The penalty for non-compliance is only ₹25,000, so there is little money behind the problem either.

8. If we want a third option — how to engineer the missing novelty

Three ideas have working data and fail only on novelty. Novelty can be designed in:

PS No.	Idea	How to make it genuinely new
SIH26001	Landslide warning, NER	GSI already predicts *slopes failing*. Nobody connects that to **road severance** — which village loses its only access route, and for how long. Different output, different mathematics (graph connectivity, not slope physics).
SIH26066	OceanEmbed	Published work builds global models. An **Indian-Ocean-specific model validated against INCOIS moorings**, with honest error bars by depth, is a contribution MoES judges would recognise.
SIH26162	Industrial fires	NASA FIRMS tells you *a fire exists*. Linking persistent heat sources to **registered industrial units** changes the question to "which plant is burning continuously, and is it permitted?"

Note on SIH26001 (landslide): it scores highest on impact of all eight, but its data access is **not yet verified** — GSI's Bhukosh portal did not respond during testing. **Verify before committing to it.**

9. Data access — what was actually tested

PS No.	Data source	Result	Evidence
SIH26011	MahaRERA public portal	WORKS	Real project pulled; land shares computed for 232 units, zero error
SIH26196	CDSCO NSQ database	WORKS	Live database 2019–2026; alert PDF downloaded and parsed; batch numbers present
SIH26066	Argo floats + NOAA sea-surface temp	WORKS	Argo index downloaded live (timestamped 24 Aug 2026); NOAA file valid
SIH26162	NASA FIRMS fire feed	WORKS	1,046 live detections, no API key needed, includes fire radiative power
SIH26100	GeM public bid listing	WORKS	48,266 bids listed; a real bid document downloaded and read
SIH26001	GSI Bhukosh / NASA rainfall	UNVERIFIED	Portal timed out twice; recorded honestly rather than assumed working
SIH26102	MPLADS eSAKSHI	FAILS	Allocation data only — no work-level records exist
SIH26034	Product label images	FAILS	No public dataset exists anywhere

10. What the team needs to decide

1. **Pick one primary idea.** Recommendation: **SIH26011**. It is the only one whose core calculation has been demonstrated working on real government data.
2. **Pick a backup.** Recommendation: **drug batch alerts** — easiest to build, clearest one-line pitch, eight years of ready data.
3. **Confirm with the college SPOC:** SIH requires an internal college hackathon to shortlist teams before portal registration, and each team may submit at most **two ideas**. Idea submission closes **20 September 2026**.
4. **If a third is wanted**, verify SIH26001's data access first — highest impact of all eight, but unproven access.

Why we are handing over 2 strong ideas rather than 20

A longer list was achievable by relaxing the bar — but the extra entries would be ideas a judge dismantles with one question. Every score here was set *after* checking competitors and *after* opening the actual data source. Three of our earlier front-runners were killed at exactly this stage, which is the point: better to lose an idea in August than in the finals.

Full working notes, every rejected problem statement with its reason, and the complete 226-PS dataset: `~/dev/sih2026/`